

TestAmerica

THE LEADER IN ENVIRONMENTAL TESTING

ANALYTICAL REPORT

TestAmerica Laboratories, Inc.

TestAmerica Sacramento

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West Sacramento, CA 95605

Tel: (916)373-5600

TestAmerica Job ID: 320-4452-1

Client Project/Site: Stringfellow Acid Project

For:

CH2M Hill Constructors, Inc.

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Suite 200

Santa Ana, California 92707

Attn: Emily Keene



Authorized for release by:

10/31/2013 2:00:19 PM

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Results relate only to the items tested and the sample(s) as received by the laboratory.

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Definitions/Glossary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Qualifiers

GC/MS VOA

Qualifier	Qualifier Description
J	Result is less than the RL but greater than or equal to the MDL and the concentration is an approximate value.

GC Semi VOA

Qualifier	Qualifier Description
B	Compound was found in the blank and sample.
J	Result is less than the RL but greater than or equal to the MDL and the concentration is an approximate value.
X	Surrogate is outside control limits

HPLC/IC

Qualifier	Qualifier Description
J	Result is less than the RL but greater than or equal to the MDL and the concentration is an approximate value.

LCMS

Qualifier	Qualifier Description
F	MS/MSD Recovery and/or RPD exceeds the control limits
*	Isotope Dilution analyte exceeds control limits
*	LCS or LCSD exceeds the control limits
J	Result is less than the RL but greater than or equal to the MDL and the concentration is an approximate value.
E	Result exceeded calibration range.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
□	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CNF	Contains no Free Liquid
DER	Duplicate error ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision level concentration
MDA	Minimum detectable activity
EDL	Estimated Detection Limit
MDC	Minimum detectable concentration
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative error ratio
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

Case Narrative

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Job ID: 320-4452-1

Laboratory: TestAmerica Sacramento

Narrative

Job Narrative 320-4452-1

Receipt

The samples were received on 10/8/2013 8:50 AM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperatures of the 2 coolers at receipt time were 1.7° C and 3.2° C.

Except:

The container label for the following sample(s) did not match the information listed on the Chain-of-Custody (COC): . Did not receive EPAZ531CS03.

GC/MS VOA - GRO

Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with batch 27338.

The closing continuing calibration verification (CCV) for GRO (C7-C12) associated with analytical batch 27338 recovered above the upper control limit. The samples associated with this CCV were non-detect for the affected analyte. The samples were re-analyzed.

The re-analysis sequence (batch 28211) also had a closing gas CCV that recovered above the upper acceptance limit. The method blank and some samples had positive detections of the target analyte below the reporting limit. It was determined that the instrument sensitivity had drifted upwards after the start of the analysis sequence, resulting in a high bias for client and quality control samples. Sample 320-4452-6 was re-analyzed outside of the analytical holding time. There was no more sample volume for re-analysis. Both sets of data are reported. The original sample injections are reported as the primary result; the re-analysis injections are reported as the secondary results.

The original sample injection data is considered more technically sound. The high-biased CCV in the original run sequence was clearly due to carry over from the previous sample injection which had visible high levels of hydrocarbons near the back end of the C7-C12 GRO window. The affected samples were all analyzed prior to the sample with severe matrix. There was an additional GRO CCV analyzed after the failing CCV, that did not bracket these samples, and did meet acceptance limits for the C7-C12 gasoline range, indicating the system had returned to control.

The internal standard, 1,4-difluorobenzene, had a response that was just below the lower acceptance limit for sample EPAZ5S31AS03 (320-4452-2). The sample was re-analyzed on 10/18/13 with confirming results. The original injection of the sample is reported. The surrogate recovery is in control.

The internal standard, 1,4-difluorobenzene, had a response below the lower acceptance limit in sample EPAZ5S31CS08 (320-4452-8). Re-analysis of the sample was performed that yielded similar results. The original injection is reported.

Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with analytical batch 27878.

Reanalysis of the following sample was performed outside of the analytical holding time: EPAZ5S31BS3 (320-4452-6).

Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with analytical batch 28211.

No other analytical or quality issues were noted.

GC/MS Semi VOA - NDMA

The following samples and associated MB and LCS were not spiked with recovery standard EPAZ5S31AS00 (320-4452-1), EPAZ5S31AS03 (320-4452-2), EPAZ5S31BS00 (320-4452-3), EPAZ5S31BS03 (320-4452-4), EPAZ5S31BS08 (320-4452-5), EPAZ5S31CS00 (320-4452-7), EPAZ5S31CS08 (320-4452-8), EPAZ5S31DS00 (320-4452-9), EPAZ5S31DS03 (320-4452-10), EPAZ5S31DS53 (320-4452-11), EPAZ5S31DS53 (320-4452-11 MS), EPAZ5S31DS53 (320-4452-11 MSD). This error was discovered upon analysis of the samples. The appropriate amount of recovery standard was added and the sample reconcentrated. The sample was re-injected and results reported. The recovery of the isotope dilution analytes were within acceptance criteria.

Case Narrative

Client: CH2M Hill Constructors, Inc.
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Job ID: 320-4452-1 (Continued)

Laboratory: TestAmerica Sacramento (Continued)

The matrix spike / matrix spike duplicate (MS/MSD) percent recoveries and %RPD for prep batch 27333 (analytical batch 27814) were outside control limits. This is attributed to: non-homogeneity of the sample matrix. The associated laboratory control sample (LCS) recoveries met acceptance criteria.

The isotope dilution analyte (IDA) recovery for the following samples were outside acceptance limits: EPAZ5S31AS00 (320-4452-1), EPAZ5S31BS03 (320-4452-4). No corrective action was performed per the laboratory's standard operating procedure since the recovery was above 10% and the signal to noise criteria was met.

The laboratory control sample (LCS), matrix spike (MS) and matrix spike duplicate (MSD) for prep batch 27542 (analytical batch 28138) recovered outside control limits for the following analytes: NDMA. These analytes were biased high in the LCS, MS and MSD and were not detected in the associated samples. Therefore, the data have been reported.

The isotope dilution analyte (IDA) recovery for the following samples were outside acceptance limits: EPAZ5S31DS53 (320-4452-11 MS), EPAZ5S31DS53 (320-4452-11 MSD). No corrective action was performed per the laboratory's standard operating procedure since the recovery was above 10% and the signal to noise criteria was met.

The laboratory control sample (LCS), matrix spike (MS) and matrix spike duplicate (MSD) for prep batch 27642 (analytical batch 28194) recovered outside control limits for the following analytes: NDMA. These analytes were biased high in the LCS, MS, and MSD and were not detected in the associated samples. Therefore, the data have been reported. Additionally, for sample 4452-11 MSD, recoveries were over the calibration range and is being reported with an E flag.

No other analytical or quality issues were noted.

HPLC

No analytical or quality issues were noted.

GC Semi VOA -DRO

The surrogate recovery for the following samples was outside control limits: EPAZ5S31CS00 (320-4452-7). The samples indicates evidence of matrix interference; therefore, re-extraction and/or re-analysis was not performed.

Insufficient sample volume was available to perform a matrix spike/matrix spike duplicate (MS/MSD) associated with batch 27435 for method 8015B_DRO-aqueous.

The laboratory control sample (LCS) associated with samples EPAZ5S31AS00 (320-4452-1), EPAZ5S31AS03 (320-4452-2), EPAZ5S31BS00 (320-4452-3), EPAZ5S31BS03 (320-4452-4), EPAZ5S31BS08 (320-4452-5), EPAZ5S31CS00 (320-4452-7), EPAZ5S31CS08 (320-4452-8), EPAZ5S31DS00 (320-4452-9), EPAZ5S31DS03 (320-4452-10), EPAZ5S31DS53 (320-4452-11), EPAZ5S31DS53 (320-4452-11 MS), EPAZ5S31DS53 (320-4452-11 MSD) for method 8015-solid was concentrated to approximately 4mL instead of the recommended 8mL on the concentration bath prior to brining the extract to final volume under nitrogen. This should not affect surrogate or analyte recoveries.

No other analytical or quality issues were noted.

LCMS

No analytical or quality issues were noted.

General Chemistry - Method 9056

Method(s) 9056: The sulfate results for the following samples: (320-4452-11 DU), EPAZ5S31DS53 (320-4452-11), are less than one half the reporting limit. Because these results are estimated values the RPD limits do not apply for these samples.

Method(s) 9056: The nitrate results for the following samples: (320-4452-11 DU), EPAZ5S31DS53 (320-4452-11), are less than the reporting limit and flagged as J values. As a result the RPD limits are not applicable for the sample and it's duplicate.

No other analytical or quality issues were noted.

I certify that this data package is in compliance with the terms and conditions of the contract and Statement of Work both technically and for

Case Narrative

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

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Laboratory: TestAmerica Sacramento (Continued)

completeness, for other than the condition(s) detailed above. Release of the data contained in this hardcopy data package and in the computer-readable data submitted has been authorized by the Quality Assurance Manager or his/her designee, as verified by the following signature.

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Detection Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31AS00

Lab Sample ID: 320-4452-1

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	4.6	B	1.1	0.32	mg/Kg	1	✱	8015B	Total/NA
Fluoride	1.1	J	2.7	0.32	mg/Kg	1	✱	9056	Soluble
Nitrate	1.7		0.27	0.12	mg/Kg	1	✱	9056	Soluble
Sulfate	4.9	J	5.3	0.26	mg/Kg	1	✱	9056	Soluble
Perchlorate	1.0		1.0	0.16	ug/Kg	1	✱	6850	Total/NA

Client Sample ID: EPAZ5S31AS03

Lab Sample ID: 320-4452-2

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	1.5	B	1.0	0.31	mg/Kg	1	✱	8015B	Total/NA
Fluoride	0.66	J	2.6	0.31	mg/Kg	1	✱	9056	Soluble
Nitrate	0.98		0.26	0.11	mg/Kg	1	✱	9056	Soluble
Sulfate	1.5	J	5.2	0.25	mg/Kg	1	✱	9056	Soluble
Perchlorate	0.20	J	1.0	0.15	ug/Kg	1	✱	6850	Total/NA

Client Sample ID: EPAZ5S31BS00

Lab Sample ID: 320-4452-3

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Gasoline Range Organics (GRO)	0.070	J	0.58	0.058	mg/Kg	1	✱	8260B/CA_LUFT	Total/NA
-C7-C12								MS	
Diesel Range Organics (C10-C24)	1.3	B	1.0	0.30	mg/Kg	1	✱	8015B	Total/NA
Fluoride	0.83	J	2.5	0.30	mg/Kg	1	✱	9056	Soluble
Nitrate	0.39		0.25	0.11	mg/Kg	1	✱	9056	Soluble
Sulfate	4.0	J	5.1	0.24	mg/Kg	1	✱	9056	Soluble
Perchlorate	2.8		1.0	0.15	ug/Kg	1	✱	6850	Total/NA

Client Sample ID: EPAZ5S31BS03

Lab Sample ID: 320-4452-4

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Gasoline Range Organics (GRO)	0.078	J	0.55	0.055	mg/Kg	1	✱	8260B/CA_LUFT	Total/NA
-C7-C12								MS	
Diesel Range Organics (C10-C24)	2.1	B	1.0	0.30	mg/Kg	1	✱	8015B	Total/NA
Fluoride	1.1	J	2.5	0.30	mg/Kg	1	✱	9056	Soluble
Nitrate	1.7		0.25	0.11	mg/Kg	1	✱	9056	Soluble
Sulfate	2.3	J	5.1	0.24	mg/Kg	1	✱	9056	Soluble
Perchlorate	0.36	J	0.98	0.15	ug/Kg	1	✱	6850	Total/NA

Client Sample ID: EPAZ5S31BS08

Lab Sample ID: 320-4452-5

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Gasoline Range Organics (GRO)	0.050	J	0.47	0.047	mg/Kg	1	✱	8260B/CA_LUFT	Total/NA
-C7-C12								MS	
Diesel Range Organics (C10-C24)	0.52	J B	1.0	0.31	mg/Kg	1	✱	8015B	Total/NA
Fluoride	0.96	J	2.5	0.30	mg/Kg	1	✱	9056	Soluble
Nitrate	0.65		0.25	0.11	mg/Kg	1	✱	9056	Soluble
Sulfate	3.4	J	5.1	0.24	mg/Kg	1	✱	9056	Soluble
Perchlorate	0.63	J	1.0	0.15	ug/Kg	1	✱	6850	Total/NA

Client Sample ID: EPAZ5S31BS3

Lab Sample ID: 320-4452-6

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
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This Detection Summary does not include radiochemical test results.

TestAmerica Sacramento

Detection Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS3 (Continued)

Lab Sample ID: 320-4452-6

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics [C10-C28]	24	J B	48	15	ug/L	1		8015B	Total/NA

Client Sample ID: EPAZ5S31CS00

Lab Sample ID: 320-4452-7

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	16	B	1.0	0.30	mg/Kg	1	☼	8015B	Total/NA
Fluoride	1.4	J	2.5	0.30	mg/Kg	1	☼	9056	Soluble
Nitrate	1.0		0.25	0.11	mg/Kg	1	☼	9056	Soluble
Sulfate	3.6	J	5.0	0.24	mg/Kg	1	☼	9056	Soluble
Perchlorate	0.67	J	0.99	0.15	ug/Kg	1	☼	6850	Total/NA

Client Sample ID: EPAZ5S31CS08

Lab Sample ID: 320-4452-8

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	3.9	B	1.0	0.31	mg/Kg	1	☼	8015B	Total/NA
Fluoride	0.53	J	2.5	0.30	mg/Kg	1	☼	9056	Soluble
Nitrate	2.1		0.25	0.11	mg/Kg	1	☼	9056	Soluble
Sulfate	2.8	J	5.0	0.24	mg/Kg	1	☼	9056	Soluble
Perchlorate	0.21	J	1.0	0.15	ug/Kg	1	☼	6850	Total/NA

Client Sample ID: EPAZ5S31DS00

Lab Sample ID: 320-4452-9

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	1.5	B	1.0	0.31	mg/Kg	1	☼	8015B	Total/NA
Fluoride	0.78	J	2.5	0.30	mg/Kg	1	☼	9056	Soluble
Nitrate	0.45		0.25	0.11	mg/Kg	1	☼	9056	Soluble
Sulfate	2.4	J	5.0	0.24	mg/Kg	1	☼	9056	Soluble
Perchlorate	0.68	J	1.0	0.15	ug/Kg	1	☼	6850	Total/NA

Client Sample ID: EPAZ5S31DS03

Lab Sample ID: 320-4452-10

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Diesel Range Organics (C10-C24)	0.93	J B	1.0	0.31	mg/Kg	1	☼	8015B	Total/NA
Fluoride	0.90	J	2.5	0.30	mg/Kg	1	☼	9056	Soluble
Nitrate	1.2		0.25	0.11	mg/Kg	1	☼	9056	Soluble
Sulfate	6.5		5.0	0.24	mg/Kg	1	☼	9056	Soluble
Perchlorate	0.29	J	1.0	0.15	ug/Kg	1	☼	6850	Total/NA

Client Sample ID: EPAZ5S31DS53

Lab Sample ID: 320-4452-11

Analyte	Result	Qualifier	RL	MDL	Unit	Dil Fac	D	Method	Prep Type
Gasoline Range Organics (GRO)	0.055	J	0.47	0.047	mg/Kg	1	☼	8260B/CA_LUFT	Total/NA
-C7-C12								MS	
Diesel Range Organics (C10-C24)	0.61	J B	1.0	0.31	mg/Kg	1	☼	8015B	Total/NA
Fluoride	1.9	J	2.6	0.31	mg/Kg	1	☼	9056	Soluble
Nitrate	0.21	J	0.26	0.11	mg/Kg	1	☼	9056	Soluble
Sulfate	1.0	J	5.1	0.25	mg/Kg	1	☼	9056	Soluble

This Detection Summary does not include radiochemical test results.

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31AS00

Lab Sample ID: 320-4452-1

Date Collected: 10/07/13 08:50

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 93.4

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		0.60	0.060	mg/Kg	☼	10/08/13 17:53	10/18/13 15:13	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	89		70 - 131				10/08/13 17:53	10/18/13 15:13	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	4.6	B	1.1	0.32	mg/Kg	☼	10/15/13 11:23	10/21/13 16:47	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	120		63 - 141				10/15/13 11:23	10/21/13 16:47	1

Method: 9056 - Anions, Ion Chromatography - Soluble

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	1.1	J	2.7	0.32	mg/Kg	☼		10/24/13 19:11	1
Nitrate	1.7		0.27	0.12	mg/Kg	☼		10/24/13 19:11	1
Sulfate	4.9	J	5.3	0.26	mg/Kg	☼		10/24/13 19:11	1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	1.0		1.0	0.16	ug/Kg	☼	10/17/13 06:52	10/21/13 21:24	1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		11	0.93	ug/Kg	☼	10/16/13 13:43	10/26/13 00:22	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	86		70 - 130				10/16/13 13:43	10/26/13 00:22	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	5.8	0.89	ug/Kg	☼	10/15/13 10:04	10/23/13 06:26	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	39	*	50 - 150				10/15/13 10:04	10/23/13 06:26	1

Client Sample ID: EPAZ5S31AS03

Lab Sample ID: 320-4452-2

Date Collected: 10/07/13 08:55

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		0.47	0.047	mg/Kg	☼	10/08/13 17:53	10/15/13 02:03	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	91		70 - 131				10/08/13 17:53	10/15/13 02:03	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	1.5	B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 17:16	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31AS03

Lab Sample ID: 320-4452-2

Date Collected: 10/07/13 08:55

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
<i>o</i> -Terphenyl (Surr)	105		63 - 141				10/15/13 11:23	10/21/13 17:16	1
Method: 9056 - Anions, Ion Chromatography - Soluble									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.66	J	2.6	0.31	mg/Kg	☼		10/24/13 19:22	1
Nitrate	0.98		0.26	0.11	mg/Kg	☼		10/24/13 19:22	1
Sulfate	1.5	J	5.2	0.25	mg/Kg	☼		10/24/13 19:22	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.20	J	1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 21:41	1
Method: 8321A - Benzene Sulfonic Acids (LCMS)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.91	ug/Kg	☼	10/16/13 13:43	10/26/13 00:40	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	72		70 - 130				10/16/13 13:43	10/26/13 00:40	1
Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	4.7	0.72	ug/Kg	☼	10/15/13 10:04	10/23/13 06:49	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	70		50 - 150				10/15/13 10:04	10/23/13 06:49	1

Client Sample ID: EPAZ5S31BS00

Lab Sample ID: 320-4452-3

Date Collected: 10/07/13 11:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.7

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	0.070	J	0.58	0.058	mg/Kg	☼	10/08/13 17:53	10/15/13 02:27	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	88		70 - 131				10/08/13 17:53	10/15/13 02:27	1
Method: 8015B - Diesel Range Organics (DRO) (GC)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	1.3	B	1.0	0.30	mg/Kg	☼	10/15/13 11:23	10/21/13 17:45	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
<i>o</i> -Terphenyl (Surr)	108		63 - 141				10/15/13 11:23	10/21/13 17:45	1
Method: 9056 - Anions, Ion Chromatography - Soluble									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.83	J	2.5	0.30	mg/Kg	☼		10/24/13 19:34	1
Nitrate	0.39		0.25	0.11	mg/Kg	☼		10/24/13 19:34	1
Sulfate	4.0	J	5.1	0.24	mg/Kg	☼		10/24/13 19:34	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	2.8		1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 21:58	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS00

Lab Sample ID: 320-4452-3

Date Collected: 10/07/13 11:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.7

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.89	ug/Kg	☼	10/16/13 13:43	10/26/13 00:57	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	95		70 - 130				10/16/13 13:43	10/26/13 00:57	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	5.4	0.82	ug/Kg	☼	10/15/13 10:04	10/23/13 07:13	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	76		50 - 150				10/15/13 10:04	10/23/13 07:13	1

Client Sample ID: EPAZ5S31BS03

Lab Sample ID: 320-4452-4

Date Collected: 10/07/13 11:10

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.4

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	0.078	J	0.55	0.055	mg/Kg	☼	10/08/13 17:53	10/18/13 16:00	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	89		70 - 131				10/08/13 17:53	10/18/13 16:00	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	2.1	B	1.0	0.30	mg/Kg	☼	10/15/13 11:23	10/21/13 18:14	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	104		63 - 141				10/15/13 11:23	10/21/13 18:14	1

Method: 9056 - Anions, Ion Chromatography - Soluble

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	1.1	J	2.5	0.30	mg/Kg	☼		10/24/13 19:45	1
Nitrate	1.7		0.25	0.11	mg/Kg	☼		10/24/13 19:45	1
Sulfate	2.3	J	5.1	0.24	mg/Kg	☼		10/24/13 19:45	1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.36	J	0.98	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 22:14	1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.89	ug/Kg	☼	10/16/13 13:43	10/26/13 01:15	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	92		70 - 130				10/16/13 13:43	10/26/13 01:15	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	4.9	0.74	ug/Kg	☼	10/15/13 10:04	10/23/13 07:37	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	46	*	50 - 150				10/15/13 10:04	10/23/13 07:37	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS08

Lab Sample ID: 320-4452-5

Date Collected: 10/07/13 11:35

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 97.7

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	0.050	J	0.47	0.047	mg/Kg	☼	10/08/13 17:53	10/18/13 16:23	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	89		70 - 131				10/08/13 17:53	10/18/13 16:23	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	0.52	J B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 18:43	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	101		63 - 141				10/15/13 11:23	10/21/13 18:43	1

Method: 9056 - Anions, Ion Chromatography - Soluble

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.96	J	2.5	0.30	mg/Kg	☼		10/24/13 19:57	1
Nitrate	0.65		0.25	0.11	mg/Kg	☼		10/24/13 19:57	1
Sulfate	3.4	J	5.1	0.24	mg/Kg	☼		10/24/13 19:57	1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.63	J	1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 22:31	1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.89	ug/Kg	☼	10/16/13 13:43	10/26/13 01:32	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	81		70 - 130				10/16/13 13:43	10/26/13 01:32	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	5.0	0.76	ug/Kg	☼	10/15/13 10:04	10/23/13 08:00	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	59		50 - 150				10/15/13 10:04	10/23/13 08:00	1

Client Sample ID: EPAZ5S31BS3

Lab Sample ID: 320-4452-6

Date Collected: 10/07/13 13:30

Matrix: Water

Date Received: 10/08/13 08:50

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		50	15	ug/L	—		10/11/13 20:55	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	97		73 - 115					10/11/13 20:55	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics [C10-C28]	24	J B	48	15	ug/L	—	10/14/13 10:08	10/22/13 06:20	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS3

Lab Sample ID: 320-4452-6

Date Collected: 10/07/13 13:30

Matrix: Water

Date Received: 10/08/13 08:50

Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
<i>o</i> -Terphenyl (Surr)	96		56 - 145				10/14/13 10:08	10/22/13 06:20	1
Method: 9056 - Anions, Ion Chromatography									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	ND		0.50	0.059	mg/L			10/08/13 14:08	1
Nitrate	ND		0.050	0.022	mg/L			10/08/13 14:08	1
Sulfate	ND		1.0	0.049	mg/L			10/08/13 14:08	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	ND		0.50	0.082	ug/L		10/16/13 07:14	10/19/13 09:01	1
Method: 8321A - Benzene Sulfonic Acids (LCMS)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		1.0	0.097	ug/L		10/17/13 08:50	10/25/13 21:44	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	102		63 - 123				10/17/13 08:50	10/25/13 21:44	1
Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND		1.9	0.37	ng/L		10/11/13 12:19	10/18/13 06:25	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	44		25 - 150				10/11/13 12:19	10/18/13 06:25	1

Client Sample ID: EPAZ5S31CS00

Lab Sample ID: 320-4452-7

Date Collected: 10/07/13 10:15

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		0.56	0.056	mg/Kg	☼	10/08/13 17:53	10/18/13 16:47	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	90		70 - 131				10/08/13 17:53	10/18/13 16:47	1
Method: 8015B - Diesel Range Organics (DRO) (GC)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	16	B	1.0	0.30	mg/Kg	☼	10/15/13 11:23	10/21/13 19:12	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	181	X	63 - 141				10/15/13 11:23	10/21/13 19:12	1
Method: 9056 - Anions, Ion Chromatography - Soluble									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	1.4	J	2.5	0.30	mg/Kg	☼		10/24/13 20:08	1
Nitrate	1.0		0.25	0.11	mg/Kg	☼		10/24/13 20:08	1
Sulfate	3.6	J	5.0	0.24	mg/Kg	☼		10/24/13 20:08	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.67	J	0.99	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 22:48	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31CS00

Lab Sample ID: 320-4452-7

Date Collected: 10/07/13 10:15

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.88	ug/Kg	☼	10/16/13 13:43	10/26/13 01:50	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	93		70 - 130				10/16/13 13:43	10/26/13 01:50	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	5.2	0.79	ug/Kg	☼	10/15/13 10:04	10/23/13 08:24	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	76		50 - 150				10/15/13 10:04	10/23/13 08:24	1

Client Sample ID: EPAZ5S31CS08

Lab Sample ID: 320-4452-8

Date Collected: 10/07/13 10:55

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.8

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO)	ND		0.49	0.049	mg/Kg	☼	10/08/13 17:53	10/18/13 17:10	1
-C7-C12									
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	88		70 - 131				10/08/13 17:53	10/18/13 17:10	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	3.9	B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 19:41	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	114		63 - 141				10/15/13 11:23	10/21/13 19:41	1

Method: 9056 - Anions, Ion Chromatography - Soluble

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.53	J	2.5	0.30	mg/Kg	☼		10/24/13 20:20	1
Nitrate	2.1		0.25	0.11	mg/Kg	☼		10/24/13 20:20	1
Sulfate	2.8	J	5.0	0.24	mg/Kg	☼		10/24/13 20:20	1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.21	J	1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 23:05	1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.89	ug/Kg	☼	10/16/13 13:43	10/26/13 02:08	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	89		70 - 130				10/16/13 13:43	10/26/13 02:08	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	4.7	0.72	ug/Kg	☼	10/15/13 10:04	10/23/13 08:51	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	89		50 - 150				10/15/13 10:04	10/23/13 08:51	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31DS00

Lab Sample ID: 320-4452-9

Date Collected: 10/07/13 07:20

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		0.51	0.051	mg/Kg	☼	10/08/13 17:53	10/18/13 17:33	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	91		70 - 131				10/08/13 17:53	10/18/13 17:33	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	1.5	B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 20:10	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	116		63 - 141				10/15/13 11:23	10/21/13 20:10	1

Method: 9056 - Anions, Ion Chromatography - Soluble

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.78	J	2.5	0.30	mg/Kg	☼		10/24/13 20:31	1
Nitrate	0.45		0.25	0.11	mg/Kg	☼		10/24/13 20:31	1
Sulfate	2.4	J	5.0	0.24	mg/Kg	☼		10/24/13 20:31	1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.68	J	1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 23:21	1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.88	ug/Kg	☼	10/16/13 13:43	10/26/13 02:25	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	89		70 - 130				10/16/13 13:43	10/26/13 02:25	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	5.2	0.79	ug/Kg	☼	10/15/13 10:04	10/23/13 13:33	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	62		50 - 150				10/15/13 10:04	10/23/13 13:33	1

Client Sample ID: EPAZ5S31DS03

Lab Sample ID: 320-4452-10

Date Collected: 10/07/13 08:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.3

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		0.52	0.052	mg/Kg	☼	10/08/13 17:53	10/18/13 17:57	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	92		70 - 131				10/08/13 17:53	10/18/13 17:57	1

Method: 8015B - Diesel Range Organics (DRO) (GC)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	0.93	J B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 22:07	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31DS03

Lab Sample ID: 320-4452-10

Date Collected: 10/07/13 08:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.3

Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
<i>o</i> -Terphenyl (Surr)	103		63 - 141				10/15/13 11:23	10/21/13 22:07	1
Method: 9056 - Anions, Ion Chromatography - Soluble									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	0.90	J	2.5	0.30	mg/Kg	☼		10/24/13 21:06	1
Nitrate	1.2		0.25	0.11	mg/Kg	☼		10/24/13 21:06	1
Sulfate	6.5		5.0	0.24	mg/Kg	☼		10/24/13 21:06	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	0.29	J	1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 23:38	1
Method: 8321A - Benzene Sulfonic Acids (LCMS)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.88	ug/Kg	☼	10/16/13 13:43	10/26/13 03:00	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	83		70 - 130				10/16/13 13:43	10/26/13 03:00	1
Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	4.7	0.71	ug/Kg	☼	10/15/13 10:04	10/23/13 13:56	1
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	61		50 - 150				10/15/13 10:04	10/23/13 13:56	1

Client Sample ID: EPAZ5S31DS53

Lab Sample ID: 320-4452-11

Date Collected: 10/07/13 08:30

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	0.055	J	0.47	0.047	mg/Kg	☼	10/08/13 17:53	10/15/13 01:17	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	92		70 - 131				10/08/13 17:53	10/15/13 01:17	1
Method: 8015B - Diesel Range Organics (DRO) (GC)									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	0.61	J B	1.0	0.31	mg/Kg	☼	10/15/13 11:23	10/21/13 22:36	1
Surrogate	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
<i>o</i> -Terphenyl (Surr)	96		63 - 141				10/15/13 11:23	10/21/13 22:36	1
Method: 9056 - Anions, Ion Chromatography - Soluble									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	1.9	J	2.6	0.31	mg/Kg	☼		10/24/13 21:17	1
Nitrate	0.21	J	0.26	0.11	mg/Kg	☼		10/24/13 21:17	1
Sulfate	1.0	J	5.1	0.25	mg/Kg	☼		10/24/13 21:17	1
Method: 6850 - Perchlorate by LC/MS or LC/MS/MS									
Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	ND		1.0	0.15	ug/Kg	☼	10/17/13 06:52	10/21/13 23:55	1

TestAmerica Sacramento

Client Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31DS53

Lab Sample ID: 320-4452-11

Date Collected: 10/07/13 08:30

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.91	ug/Kg	☼	10/16/13 13:43	10/26/13 03:18	1

Surrogate	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	82		70 - 130	10/16/13 13:43	10/26/13 03:18	1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Analyte	Result	Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND	*	4.9	0.75	ug/Kg	☼	10/15/13 10:04	10/23/13 14:20	1

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	62		50 - 150	10/15/13 10:04	10/23/13 14:20	1

Surrogate Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Matrix: Solid

Prep Type: Total/NA

		Percent Surrogate Recovery (Acceptance Limits)					
Lab Sample ID	Client Sample ID	BFB (70-131)					
320-4452-1	EPAZ5S31AS00	89					
320-4452-2	EPAZ5S31AS03	91					
320-4452-3	EPAZ5S31BS00	88					
320-4452-4	EPAZ5S31BS03	89					
320-4452-5	EPAZ5S31BS08	89					
320-4452-7	EPAZ5S31CS00	90					
320-4452-8	EPAZ5S31CS08	88					
320-4452-9	EPAZ5S31DS00	91					
320-4452-10	EPAZ5S31DS03	92					
320-4452-11	EPAZ5S31DS53	92					
320-4452-11 MS	EPAZ5S31DS53	91					
320-4452-11 MSD	EPAZ5S31DS53	91					
LCS 320-27514/15	Lab Control Sample	94					
LCS 320-27878/9	Lab Control Sample	90					
MB 320-27514/23	Method Blank	93					
MB 320-27878/11	Method Blank	91					
Surrogate Legend							
BFB = 4-Bromofluorobenzene (Surr)							

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Matrix: Water

Prep Type: Total/NA

		Percent Surrogate Recovery (Acceptance Limits)					
Lab Sample ID	Client Sample ID	BFB (73-115)					
320-4452-6	EPAZ5S31BS3	97					
320-4452-6 - RA	EPAZ5S31BS3	100					
LCS 320-27338/9	Lab Control Sample	95					
LCS 320-28211/11	Lab Control Sample	77					
LCSD 320-27338/10	Lab Control Sample Dup	94					
MB 320-27338/18	Method Blank	94					
MB 320-28211/16	Method Blank	98					
Surrogate Legend							
BFB = 4-Bromofluorobenzene (Surr)							

Method: 8015B - Diesel Range Organics (DRO) (GC)

Matrix: Solid

Prep Type: Total/NA

		Percent Surrogate Recovery (Acceptance Limits)					
Lab Sample ID	Client Sample ID	OTPH1 (63-141)					
320-4452-1	EPAZ5S31AS00	120					
320-4452-2	EPAZ5S31AS03	105					
320-4452-3	EPAZ5S31BS00	108					
320-4452-4	EPAZ5S31BS03	104					
320-4452-5	EPAZ5S31BS08	101					
320-4452-7	EPAZ5S31CS00	181 X					
320-4452-8	EPAZ5S31CS08	114					

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Surrogate Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8015B - Diesel Range Organics (DRO) (GC) (Continued)

Matrix: Solid

Prep Type: Total/NA

			Percent Surrogate Recovery (Acceptance Limits)				
Lab Sample ID	Client Sample ID	OTPH1 (63-141)					
320-4452-9	EPAZ5S31DS00	116					
320-4452-10	EPAZ5S31DS03	103					
320-4452-11	EPAZ5S31DS53	96					
320-4452-11 MS	EPAZ5S31DS53	103					
320-4452-11 MSD	EPAZ5S31DS53	109					
LCS 320-27563/2-A	Lab Control Sample	100					
MB 320-27563/1-A	Method Blank	101					
Surrogate Legend							
OTPH = o-Terphenyl (Surr)							

Method: 8015B - Diesel Range Organics (DRO) (GC)

Matrix: Water

Prep Type: Total/NA

			Percent Surrogate Recovery (Acceptance Limits)				
Lab Sample ID	Client Sample ID	OTPH1 (56-145)					
320-4452-6	EPAZ5S31BS3	96					
LCS 320-27435/2-A	Lab Control Sample	100					
MB 320-27435/1-A	Method Blank	91					
Surrogate Legend							
OTPH = o-Terphenyl (Surr)							

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Matrix: Solid

Prep Type: Total/NA

			Percent Surrogate Recovery (Acceptance Limits)				
Lab Sample ID	Client Sample ID	Benzenesulf (70-130)					
320-4452-1	EPAZ5S31AS00	86					
320-4452-2	EPAZ5S31AS03	72					
320-4452-3	EPAZ5S31BS00	95					
320-4452-4	EPAZ5S31BS03	92					
320-4452-5	EPAZ5S31BS08	81					
320-4452-7	EPAZ5S31CS00	93					
320-4452-8	EPAZ5S31CS08	89					
320-4452-9	EPAZ5S31DS00	89					
320-4452-10	EPAZ5S31DS03	83					
320-4452-11	EPAZ5S31DS53	82					
320-4452-11 MS	EPAZ5S31DS53	83					
320-4452-11 MSD	EPAZ5S31DS53	81					
LCS 320-27706/2-A	Lab Control Sample	100					
MB 320-27706/1-A	Method Blank	98					
Surrogate Legend							
4-Bromobenzenesulfonic Acid = 4-Bromobenzenesulfonic Acid							

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Surrogate Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Matrix: Water

Prep Type: Total/NA

		Percent Surrogate Recovery (Acceptance Limits)	
Lab Sample ID	Client Sample ID	Benzenesulfonic Acid (63-123)	
320-4452-6	EPAZ5S31BS3	102	
320-4452-6 MS	EPAZ5S31BS3	98	
320-4452-6 MSD	EPAZ5S31BS3	100	
LCS 320-27742/2-A	Lab Control Sample	104	
MB 320-27742/1-A	Method Blank	97	

Surrogate Legend

4-Bromobenzenesulfonic Acid = 4-Bromobenzenesulfonic Acid

Isotope Dilution Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Matrix: Solid

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	NDMA (50-150)
320-4421-C-3-C MSD	Matrix Spike Duplicate	53
320-4421-D-3-B MS	Matrix Spike	53
320-4452-1	EPAZ5S31AS00	39 *
320-4452-2	EPAZ5S31AS03	70
320-4452-3	EPAZ5S31BS00	76
320-4452-4	EPAZ5S31BS03	46 *
320-4452-5	EPAZ5S31BS08	59
320-4452-7	EPAZ5S31CS00	76
320-4452-8	EPAZ5S31CS08	89
320-4452-9	EPAZ5S31DS00	62
320-4452-10	EPAZ5S31DS03	61
320-4452-11	EPAZ5S31DS53	62
320-4452-11 MS	EPAZ5S31DS53	43 *
320-4452-11 MSD	EPAZ5S31DS53	38 *
LCS 320-27542/2-A	Lab Control Sample	66
MB 320-27542/1-A	Method Blank	70
MB 320-27642/1-A	Method Blank	114

Surrogate Legend

NDMA = N-Nitrosodimethylamine-d6

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	NDMA (25-150)
320-4450-A-4-A MS	Matrix Spike	18 *
320-4450-A-4-B MSD	Matrix Spike Duplicate	14 *
320-4452-6	EPAZ5S31BS3	44
LCS 320-27333/2-A	Lab Control Sample	52
MB 320-27333/1-A	Method Blank	56

Surrogate Legend

NDMA = N-Nitrosodimethylamine-d6

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 27514

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27074

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Gasoline Range Organics (GRO) -C7-C12	0.055	J	0.914	0.985		mg/Kg	☼	102	79 - 123
Surrogate	MS %Recovery	MS Qualifier	Limits						
4-Bromofluorobenzene (Surr)	91		70 - 131						

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 27514

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27074

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Gasoline Range Organics (GRO) -C7-C12	0.055	J	0.946	0.868		mg/Kg	☼	86	79 - 123	13	30
Surrogate	MSD %Recovery	MSD Qualifier	Limits								
4-Bromofluorobenzene (Surr)	91		70 - 131								

Lab Sample ID: MB 320-27338/18

Matrix: Water

Analysis Batch: 27338

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	ND		50	15	ug/L			10/11/13 19:49	1
Surrogate	MB %Recovery	MB Qualifier	Limits						
4-Bromofluorobenzene (Surr)	94		73 - 115						

Lab Sample ID: LCS 320-27338/9

Matrix: Water

Analysis Batch: 27338

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Gasoline Range Organics (GRO) -C7-C12	500	539		ug/L		108	78 - 118
Surrogate	LCS %Recovery	LCS Qualifier	Limits				
4-Bromofluorobenzene (Surr)	95		73 - 115				

Lab Sample ID: LCSD 320-27338/10

Matrix: Water

Analysis Batch: 27338

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Gasoline Range Organics (GRO) -C7-C12	500	535		ug/L		107	78 - 118	1	23

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QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS (Continued)

Lab Sample ID: LCSD 320-27338/10

Matrix: Water

Analysis Batch: 27338

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

	LCSD	LCSD	
Surrogate	%Recovery	Qualifier	Limits
4-Bromofluorobenzene (Surr)	94		73 - 115

Lab Sample ID: MB 320-27514/23

Matrix: Solid

Analysis Batch: 27514

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB	MB	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO)	ND		0.50	0.050	mg/Kg	-		10/14/13 22:34	1
-C7-C12									
Surrogate	MB	MB	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	93		70 - 131					10/14/13 22:34	1

Lab Sample ID: LCS 320-27514/15

Matrix: Solid

Analysis Batch: 27514

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike	LCS	LCS	Unit	D	%Rec	%Rec.
Gasoline Range Organics (GRO)	Added	Result	Qualifier	mg/Kg	-	102	Limits
-C7-C12	1.00	1.02					79 - 123
Surrogate	LCS	LCS	Limits				
4-Bromofluorobenzene (Surr)	%Recovery	Qualifier	70 - 131				
	94						

Lab Sample ID: MB 320-27878/11

Matrix: Solid

Analysis Batch: 27878

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB	MB	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO)	ND		0.50	0.050	mg/Kg	-		10/18/13 14:50	1
-C7-C12									
Surrogate	MB	MB	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	91		70 - 131					10/18/13 14:50	1

Lab Sample ID: LCS 320-27878/9

Matrix: Solid

Analysis Batch: 27878

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike	LCS	LCS	Unit	D	%Rec	%Rec.
Gasoline Range Organics (GRO)	Added	Result	Qualifier	mg/Kg	-	99	Limits
-C7-C12	1.00	0.995					79 - 123
Surrogate	LCS	LCS	Limits				
4-Bromofluorobenzene (Surr)	%Recovery	Qualifier	70 - 131				
	90						

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QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8260B/CA_LUFTMS - Volatile Organic Compounds by GC/MS (Continued)

Lab Sample ID: MB 320-28211/16

Matrix: Water

Analysis Batch: 28211

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Gasoline Range Organics (GRO) -C7-C12	15.0	J	50	15	ug/L	-		10/22/13 16:55	1
Surrogate	%Recovery	MB Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromofluorobenzene (Surr)	98		73 - 115					10/22/13 16:55	1

Lab Sample ID: LCS 320-28211/11

Matrix: Water

Analysis Batch: 28211

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Gasoline Range Organics (GRO) -C7-C12	500	429		ug/L	-	86	78 - 118
Surrogate	%Recovery	LCS Qualifier	Limits				
4-Bromofluorobenzene (Surr)	77		73 - 115				

Method: 8015B - Diesel Range Organics (DRO) (GC)

Lab Sample ID: MB 320-27435/1-A

Matrix: Water

Analysis Batch: 27994

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27435

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics [C10-C28]	21.3	J	50	16	ug/L	-	10/14/13 10:08	10/22/13 05:22	1
Surrogate	%Recovery	MB Qualifier	Limits				Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	91		56 - 145				10/14/13 10:08	10/22/13 05:22	1

Lab Sample ID: LCS 320-27435/2-A

Matrix: Water

Analysis Batch: 27994

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27435

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Diesel Range Organics [C10-C28]	300	320		ug/L	-	107	53 - 123
Surrogate	%Recovery	LCS Qualifier	Limits				
o-Terphenyl (Surr)	100		56 - 145				

Lab Sample ID: MB 320-27563/1-A

Matrix: Solid

Analysis Batch: 27994

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27563

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Diesel Range Organics (C10-C24)	0.704	J	1.0	0.30	mg/Kg	-	10/15/13 11:23	10/21/13 14:51	1

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QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8015B - Diesel Range Organics (DRO) (GC) (Continued)

Lab Sample ID: MB 320-27563/1-A

Matrix: Solid

Analysis Batch: 27994

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27563

Surrogate	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
o-Terphenyl (Surr)	101		63 - 141	10/15/13 11:23	10/21/13 14:51	1

Lab Sample ID: LCS 320-27563/2-A

Matrix: Solid

Analysis Batch: 27994

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27563

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Diesel Range Organics (C10-C24)	10.0	11.1		mg/Kg		111	67 - 113
Surrogate	LCS %Recovery	LCS Qualifier	Limits				
o-Terphenyl (Surr)	100		63 - 141				

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 27994

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27563

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Diesel Range Organics (C10-C24)	0.61	J B	10.6	11.2		mg/Kg	✱	99	67 - 113
Surrogate	MS %Recovery	MS Qualifier	Limits						
o-Terphenyl (Surr)	103		63 - 141						

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 27994

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27563

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Diesel Range Organics (C10-C24)	0.61	J B	10.5	12.1		mg/Kg	✱	109	67 - 113	8	30
Surrogate	MSD %Recovery	MSD Qualifier	Limits								
o-Terphenyl (Surr)	109		63 - 141								

Method: 9056 - Anions, Ion Chromatography

Lab Sample ID: MB 320-28225/3

Matrix: Water

Analysis Batch: 28225

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	ND		0.50	0.059	mg/L			10/08/13 10:29	1
Sulfate	ND		1.0	0.049	mg/L			10/08/13 10:29	1

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QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 9056 - Anions, Ion Chromatography (Continued)

Lab Sample ID: LCS 320-28225/4

Matrix: Water

Analysis Batch: 28225

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Fluoride	7.50	7.49		mg/L		100	90 - 110
Sulfate	7.50	7.15		mg/L		95	90 - 110

Lab Sample ID: MB 320-28226/3

Matrix: Water

Analysis Batch: 28226

Client Sample ID: Method Blank

Prep Type: Total/NA

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Nitrate	ND		0.050	0.022	mg/L			10/08/13 10:29	1

Lab Sample ID: LCS 320-28226/4

Matrix: Water

Analysis Batch: 28226

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Nitrate	1.88	1.76		mg/L		94	90 - 110

Lab Sample ID: MB 320-28360/1-A

Matrix: Solid

Analysis Batch: 28765

Client Sample ID: Method Blank

Prep Type: Soluble

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Fluoride	ND		2.5	0.30	mg/Kg			10/24/13 18:48	1
Sulfate	ND		5.0	0.24	mg/Kg			10/24/13 18:48	1

Lab Sample ID: LCS 320-28360/2-A

Matrix: Solid

Analysis Batch: 28765

Client Sample ID: Lab Control Sample

Prep Type: Soluble

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Fluoride	25.0	24.7		mg/Kg		99	85 - 115
Sulfate	25.0	24.8		mg/Kg		99	85 - 115

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 28765

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Fluoride	1.9	J	25.4	26.3		mg/Kg	☼	96	85 - 115
Sulfate	1.0	J	25.4	25.8		mg/Kg	☼	97	85 - 115

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 28765

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Fluoride	1.9	J	25.4	25.7		mg/Kg	☼	94	85 - 115	2	15
Sulfate	1.0	J	25.4	25.5		mg/Kg	☼	96	85 - 115	1	15

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 9056 - Anions, Ion Chromatography (Continued)

Lab Sample ID: 320-4452-11 DU

Matrix: Solid

Analysis Batch: 28765

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	DU Result	DU Qualifier	Unit	D	RPD	RPD Limit
Fluoride	1.9	J	1.83	J	mg/Kg	✱	3	15
Sulfate	1.0	J	1.53	J	mg/Kg	✱	40	15

Lab Sample ID: MB 320-28360/1-A

Matrix: Solid

Analysis Batch: 28766

Client Sample ID: Method Blank

Prep Type: Soluble

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Nitrate	ND		0.25	0.11	mg/Kg			10/24/13 18:48	1

Lab Sample ID: LCS 320-28360/2-A

Matrix: Solid

Analysis Batch: 28766

Client Sample ID: Lab Control Sample

Prep Type: Soluble

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec Limits
Nitrate	5.00	5.14		mg/Kg		103	85 - 115

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 28766

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec Limits
Nitrate	0.21	J	5.08	5.42		mg/Kg	✱	102	85 - 115

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 28766

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec Limits	RPD	RPD Limit
Nitrate	0.21	J	5.08	5.31		mg/Kg	✱	100	85 - 115	2	15

Lab Sample ID: 320-4452-11 DU

Matrix: Solid

Analysis Batch: 28766

Client Sample ID: EPAZ5S31DS53

Prep Type: Soluble

Analyte	Sample Result	Sample Qualifier	DU Result	DU Qualifier	Unit	D	RPD	RPD Limit
Nitrate	0.21	J	0.239	J	mg/Kg	✱	11	15

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS

Lab Sample ID: MB 320-27626/1-A

Matrix: Water

Analysis Batch: 27864

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27626

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	ND		0.50	0.082	ug/L		10/16/13 07:14	10/19/13 05:24	1

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 6850 - Perchlorate by LC/MS or LC/MS/MS (Continued)

Lab Sample ID: LCS 320-27626/2-A

Matrix: Water

Analysis Batch: 27864

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27626

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perchlorate	5.00	5.08		ug/L		102	80 - 120

Lab Sample ID: 720-52885-A-10-B MS

Matrix: Water

Analysis Batch: 27864

Client Sample ID: Matrix Spike

Prep Type: Total/NA

Prep Batch: 27626

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Perchlorate	ND		5.00	5.10		ug/L		102	80 - 120

Lab Sample ID: 720-52885-A-10-C MSD

Matrix: Water

Analysis Batch: 27864

Client Sample ID: Matrix Spike Duplicate

Prep Type: Total/NA

Prep Batch: 27626

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perchlorate	ND		5.00	5.08		ug/L		102	80 - 120	0	15

Lab Sample ID: MB 320-27733/1-A

Matrix: Solid

Analysis Batch: 28015

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27733

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
Perchlorate	ND		1.0	0.15	ug/Kg		10/17/13 06:52	10/21/13 20:34	1

Lab Sample ID: LCS 320-27733/2-A

Matrix: Solid

Analysis Batch: 28015

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27733

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perchlorate	50.0	55.1		ug/Kg		110	80 - 120

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 28015

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27733

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
Perchlorate	ND		50.0	55.2		ug/Kg	✱	110	80 - 120

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 28015

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27733

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perchlorate	ND		49.8	54.0		ug/Kg	✱	108	80 - 120	2	15

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8321A - Benzene Sulfonic Acids (LCMS)

Lab Sample ID: MB 320-27706/1-A

Matrix: Solid

Analysis Batch: 28437

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27706

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		10	0.88	ug/Kg		10/16/13 13:43	10/25/13 23:47	1
Surrogate	%Recovery	MB Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	98		70 - 130				10/16/13 13:43	10/25/13 23:47	1

Lab Sample ID: LCS 320-27706/2-A

Matrix: Solid

Analysis Batch: 28437

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27706

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
p-CBSA	50.0	54.2		ug/Kg		108	70 - 130
Surrogate	%Recovery	LCS Qualifier	Limits				
4-Bromobenzenesulfonic Acid	100		70 - 130				

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 28437

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27706

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
p-CBSA	ND		51.4	51.3		ug/Kg	☼	100	70 - 130
Surrogate	%Recovery	MS Qualifier	Limits						
4-Bromobenzenesulfonic Acid	83		70 - 130						

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 28437

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27706

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD Limit
p-CBSA	ND		51.5	53.5		ug/Kg	☼	104	70 - 130	4 30
Surrogate	%Recovery	MSD Qualifier	Limits							
4-Bromobenzenesulfonic Acid	81		70 - 130							

Lab Sample ID: MB 320-27742/1-A

Matrix: Water

Analysis Batch: 28437

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27742

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
p-CBSA	ND		1.0	0.097	ug/L		10/17/13 08:50	10/25/13 21:09	1
Surrogate	%Recovery	MB Qualifier	Limits				Prepared	Analyzed	Dil Fac
4-Bromobenzenesulfonic Acid	97		63 - 123				10/17/13 08:50	10/25/13 21:09	1

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: 8321A - Benzene Sulfonic Acids (LCMS) (Continued)

Lab Sample ID: LCS 320-27742/2-A

Matrix: Water

Analysis Batch: 28437

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27742

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
p-CBSA	5.00	5.21		ug/L		104	60 - 127
Surrogate	%Recovery	LCS Qualifier	Limits				
4-Bromobenzenesulfonic Acid	104		63 - 123				

Lab Sample ID: 320-4452-6 MS

Matrix: Water

Analysis Batch: 28437

Client Sample ID: EPAZ5S31BS3

Prep Type: Total/NA

Prep Batch: 27742

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
p-CBSA	ND		5.00	5.29		ug/L		106	60 - 127
Surrogate	%Recovery	MS Qualifier	Limits						
4-Bromobenzenesulfonic Acid	98		63 - 123						

Lab Sample ID: 320-4452-6 MSD

Matrix: Water

Analysis Batch: 28437

Client Sample ID: EPAZ5S31BS3

Prep Type: Total/NA

Prep Batch: 27742

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
p-CBSA	ND		5.00	5.31		ug/L		106	60 - 127	0	20
Surrogate	%Recovery	MSD Qualifier	Limits								
4-Bromobenzenesulfonic Acid	100		63 - 123								

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS

Lab Sample ID: MB 320-27333/1-A

Matrix: Water

Analysis Batch: 27814

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27333

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND		2.0	0.39	ng/L		10/11/13 12:19	10/18/13 02:29	1
Isotope Dilution	%Recovery	MB Qualifier	Limits						
N-Nitrosodimethylamine-d6	56		25 - 150						

Lab Sample ID: LCS 320-27333/2-A

Matrix: Water

Analysis Batch: 27814

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27333

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
N-Nitrosodimethylamine	20.0	22.2		ng/L		111	58 - 150
Isotope Dilution	%Recovery	LCS Qualifier	Limits				
N-Nitrosodimethylamine-d6	52		25 - 150				

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS (Continued)

Lab Sample ID: 320-4450-A-4-A MS

Matrix: Water

Analysis Batch: 27814

Client Sample ID: Matrix Spike

Prep Type: Total/NA

Prep Batch: 27333

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits		
N-Nitrosodimethylamine	ND		19.0	40.7	F	ng/L		214	58 - 150		
	MS	MS									
Isotope Dilution	%Recovery	Qualifier	Limits								
N-Nitrosodimethylamine-d6	18	*	25 - 150								

Lab Sample ID: 320-4450-A-4-B MSD

Matrix: Water

Analysis Batch: 27814

Client Sample ID: Matrix Spike Duplicate

Prep Type: Total/NA

Prep Batch: 27333

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
N-Nitrosodimethylamine	ND		19.0	51.7	F	ng/L		272	58 - 150	24	20
	MSD	MSD									
Isotope Dilution	%Recovery	Qualifier	Limits								
N-Nitrosodimethylamine-d6	14	*	25 - 150								

Lab Sample ID: MB 320-27542/1-A

Matrix: Solid

Analysis Batch: 28138

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27542

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND		5.0	0.76	ug/Kg		10/15/13 10:04	10/23/13 00:56	1
	MB	MB							
Isotope Dilution	%Recovery	Qualifier	Limits				Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine-d6	70		50 - 150				10/15/13 10:04	10/23/13 00:56	1

Lab Sample ID: LCS 320-27542/2-A

Matrix: Solid

Analysis Batch: 28138

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 27542

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits		
N-Nitrosodimethylamine	5.00	6.87	*	ug/Kg		137	70 - 130		
	LCS	LCS							
Isotope Dilution	%Recovery	Qualifier	Limits						
N-Nitrosodimethylamine-d6	66		50 - 150						

Lab Sample ID: 320-4421-C-3-C MSD

Matrix: Solid

Analysis Batch: 28138

Client Sample ID: Matrix Spike Duplicate

Prep Type: Total/NA

Prep Batch: 27542

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
N-Nitrosodimethylamine	ND	*	4.63	7.71	F	ug/Kg	☼	166	70 - 130	12	35
	MSD	MSD									
Isotope Dilution	%Recovery	Qualifier	Limits								
N-Nitrosodimethylamine-d6	53		50 - 150								

TestAmerica Sacramento

QC Sample Results

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method: GCMSMS_NDMA - Nitrosamines by Isotope Dilution and GC/CI/MS/MS (Continued)

Lab Sample ID: 320-4421-D-3-B MS

Matrix: Solid

Analysis Batch: 28138

Client Sample ID: Matrix Spike

Prep Type: Total/NA

Prep Batch: 27542

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
N-Nitrosodimethylamine	ND	*	4.80	6.85	F	ug/Kg	☼	143	70 - 130
Isotope Dilution									
	%Recovery	Qualifier	Limits						
N-Nitrosodimethylamine-d6	53		50 - 150						

Lab Sample ID: 320-4452-11 MS

Matrix: Solid

Analysis Batch: 28194

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27542

Analyte	Sample Result	Sample Qualifier	Spike Added	MS Result	MS Qualifier	Unit	D	%Rec	%Rec. Limits
N-Nitrosodimethylamine	ND	*	4.96	9.55	F	ug/Kg	☼	192	70 - 130
Isotope Dilution									
	%Recovery	Qualifier	Limits						
N-Nitrosodimethylamine-d6	43	*	50 - 150						

Lab Sample ID: 320-4452-11 MSD

Matrix: Solid

Analysis Batch: 28194

Client Sample ID: EPAZ5S31DS53

Prep Type: Total/NA

Prep Batch: 27542

Analyte	Sample Result	Sample Qualifier	Spike Added	MSD Result	MSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	Limit
N-Nitrosodimethylamine	ND	*	5.02	10.2	E F	ug/Kg	☼	202	70 - 130	6	35
Isotope Dilution											
	%Recovery	Qualifier	Limits								
N-Nitrosodimethylamine-d6	38	*	50 - 150								

Lab Sample ID: MB 320-27642/1-A

Matrix: Solid

Analysis Batch: 28194

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 27642

Analyte	MB Result	MB Qualifier	RL	MDL	Unit	D	Prepared	Analyzed	Dil Fac
N-Nitrosodimethylamine	ND		5.0	0.76	ug/Kg		10/16/13 09:54	10/23/13 23:39	1
Isotope Dilution									
	%Recovery	Qualifier	Limits						
N-Nitrosodimethylamine-d6	114		50 - 150						

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

GC/MS VOA

Prep Batch: 27074

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	5035	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	5035	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	5035	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	5035	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	5035	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	5035	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	5035	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	5035	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	5035	
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	5035	
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	5035	
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	5035	

Analysis Batch: 27338

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	8260B/CA_LUFT MS	
LCS 320-27338/9	Lab Control Sample	Total/NA	Water	8260B/CA_LUFT MS	
LCSD 320-27338/10	Lab Control Sample Dup	Total/NA	Water	8260B/CA_LUFT MS	
MB 320-27338/18	Method Blank	Total/NA	Water	8260B/CA_LUFT MS	

Analysis Batch: 27514

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	8260B/CA_LUFT MS	27074
LCS 320-27514/15	Lab Control Sample	Total/NA	Solid	8260B/CA_LUFT MS	
MB 320-27514/23	Method Blank	Total/NA	Solid	8260B/CA_LUFT MS	

Analysis Batch: 27878

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	8260B/CA_LUFT MS	27074
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	8260B/CA_LUFT MS	27074

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

GC/MS VOA (Continued)

Analysis Batch: 27878 (Continued)

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	8260B/CA_LUFT	27074
LCS 320-27878/9	Lab Control Sample	Total/NA	Solid	MS	
MB 320-27878/11	Method Blank	Total/NA	Solid	8260B/CA_LUFT	
				MS	

Analysis Batch: 28211

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6 - RA	EPAZ5S31BS3	Total/NA	Water	8260B/CA_LUFT	
LCS 320-28211/11	Lab Control Sample	Total/NA	Water	MS	
MB 320-28211/16	Method Blank	Total/NA	Water	8260B/CA_LUFT	
				MS	

GC Semi VOA

Prep Batch: 27435

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	3510C	
LCS 320-27435/2-A	Lab Control Sample	Total/NA	Water	3510C	
MB 320-27435/1-A	Method Blank	Total/NA	Water	3510C	

Prep Batch: 27563

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	3550B	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	3550B	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	3550B	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	3550B	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	3550B	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	3550B	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	3550B	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	3550B	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	3550B	
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	3550B	
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	3550B	
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	3550B	
LCS 320-27563/2-A	Lab Control Sample	Total/NA	Solid	3550B	
MB 320-27563/1-A	Method Blank	Total/NA	Solid	3550B	

Analysis Batch: 27994

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	8015B	27563
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	8015B	27563
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	8015B	27563
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	8015B	27563
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	8015B	27563
320-4452-6	EPAZ5S31BS3	Total/NA	Water	8015B	27435
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	8015B	27563
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	8015B	27563
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	8015B	27563

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

GC Semi VOA (Continued)

Analysis Batch: 27994 (Continued)

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	8015B	27563
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	8015B	27563
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	8015B	27563
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	8015B	27563
LCS 320-27435/2-A	Lab Control Sample	Total/NA	Water	8015B	27435
LCS 320-27563/2-A	Lab Control Sample	Total/NA	Solid	8015B	27563
MB 320-27435/1-A	Method Blank	Total/NA	Water	8015B	27435
MB 320-27563/1-A	Method Blank	Total/NA	Solid	8015B	27563

HPLC/IC

Analysis Batch: 28225

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	9056	
LCS 320-28225/4	Lab Control Sample	Total/NA	Water	9056	
MB 320-28225/3	Method Blank	Total/NA	Water	9056	

Analysis Batch: 28226

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	9056	
LCS 320-28226/4	Lab Control Sample	Total/NA	Water	9056	
MB 320-28226/3	Method Blank	Total/NA	Water	9056	

Leach Batch: 28360

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Soluble	Solid	DI Leach	
320-4452-2	EPAZ5S31AS03	Soluble	Solid	DI Leach	
320-4452-3	EPAZ5S31BS00	Soluble	Solid	DI Leach	
320-4452-4	EPAZ5S31BS03	Soluble	Solid	DI Leach	
320-4452-5	EPAZ5S31BS08	Soluble	Solid	DI Leach	
320-4452-7	EPAZ5S31CS00	Soluble	Solid	DI Leach	
320-4452-8	EPAZ5S31CS08	Soluble	Solid	DI Leach	
320-4452-9	EPAZ5S31DS00	Soluble	Solid	DI Leach	
320-4452-10	EPAZ5S31DS03	Soluble	Solid	DI Leach	
320-4452-11	EPAZ5S31DS53	Soluble	Solid	DI Leach	
320-4452-11 DU	EPAZ5S31DS53	Soluble	Solid	DI Leach	
320-4452-11 MS	EPAZ5S31DS53	Soluble	Solid	DI Leach	
320-4452-11 MSD	EPAZ5S31DS53	Soluble	Solid	DI Leach	
LCS 320-28360/2-A	Lab Control Sample	Soluble	Solid	DI Leach	
MB 320-28360/1-A	Method Blank	Soluble	Solid	DI Leach	

Analysis Batch: 28765

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Soluble	Solid	9056	28360
320-4452-2	EPAZ5S31AS03	Soluble	Solid	9056	28360
320-4452-3	EPAZ5S31BS00	Soluble	Solid	9056	28360
320-4452-4	EPAZ5S31BS03	Soluble	Solid	9056	28360
320-4452-5	EPAZ5S31BS08	Soluble	Solid	9056	28360
320-4452-7	EPAZ5S31CS00	Soluble	Solid	9056	28360
320-4452-8	EPAZ5S31CS08	Soluble	Solid	9056	28360

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

HPLC/IC (Continued)

Analysis Batch: 28765 (Continued)

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-9	EPAZ5S31DS00	Soluble	Solid	9056	28360
320-4452-10	EPAZ5S31DS03	Soluble	Solid	9056	28360
320-4452-11	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 DU	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 MS	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 MSD	EPAZ5S31DS53	Soluble	Solid	9056	28360
LCS 320-28360/2-A	Lab Control Sample	Soluble	Solid	9056	28360
MB 320-28360/1-A	Method Blank	Soluble	Solid	9056	28360

Analysis Batch: 28766

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Soluble	Solid	9056	28360
320-4452-2	EPAZ5S31AS03	Soluble	Solid	9056	28360
320-4452-3	EPAZ5S31BS00	Soluble	Solid	9056	28360
320-4452-4	EPAZ5S31BS03	Soluble	Solid	9056	28360
320-4452-5	EPAZ5S31BS08	Soluble	Solid	9056	28360
320-4452-7	EPAZ5S31CS00	Soluble	Solid	9056	28360
320-4452-8	EPAZ5S31CS08	Soluble	Solid	9056	28360
320-4452-9	EPAZ5S31DS00	Soluble	Solid	9056	28360
320-4452-10	EPAZ5S31DS03	Soluble	Solid	9056	28360
320-4452-11	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 DU	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 MS	EPAZ5S31DS53	Soluble	Solid	9056	28360
320-4452-11 MSD	EPAZ5S31DS53	Soluble	Solid	9056	28360
LCS 320-28360/2-A	Lab Control Sample	Soluble	Solid	9056	28360
MB 320-28360/1-A	Method Blank	Soluble	Solid	9056	28360

LCMS

Prep Batch: 27333

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4450-A-4-A MS	Matrix Spike	Total/NA	Water	3510C	
320-4450-A-4-B MSD	Matrix Spike Duplicate	Total/NA	Water	3510C	
320-4452-6	EPAZ5S31BS3	Total/NA	Water	3510C	
LCS 320-27333/2-A	Lab Control Sample	Total/NA	Water	3510C	
MB 320-27333/1-A	Method Blank	Total/NA	Water	3510C	

Prep Batch: 27542

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4421-C-3-C MSD	Matrix Spike Duplicate	Total/NA	Solid	3550B	
320-4421-D-3-B MS	Matrix Spike	Total/NA	Solid	3550B	
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	3550B	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	3550B	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	3550B	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	3550B	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	3550B	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	3550B	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	3550B	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	3550B	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	3550B	

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

LCMS (Continued)

Prep Batch: 27542 (Continued)

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	3550B	
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	3550B	
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	3550B	
LCS 320-27542/2-A	Lab Control Sample	Total/NA	Solid	3550B	
MB 320-27542/1-A	Method Blank	Total/NA	Solid	3550B	

Prep Batch: 27626

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	Filtration	
720-52885-A-10-B MS	Matrix Spike	Total/NA	Water	Filtration	
720-52885-A-10-C MSD	Matrix Spike Duplicate	Total/NA	Water	Filtration	
LCS 320-27626/2-A	Lab Control Sample	Total/NA	Water	Filtration	
MB 320-27626/1-A	Method Blank	Total/NA	Water	Filtration	

Prep Batch: 27642

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
MB 320-27642/1-A	Method Blank	Total/NA	Solid	3550B	

Prep Batch: 27706

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	SHAKE	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	SHAKE	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	SHAKE	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	SHAKE	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	SHAKE	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	SHAKE	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	SHAKE	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	SHAKE	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	SHAKE	
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
LCS 320-27706/2-A	Lab Control Sample	Total/NA	Solid	SHAKE	
MB 320-27706/1-A	Method Blank	Total/NA	Solid	SHAKE	

Prep Batch: 27733

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	SHAKE	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	SHAKE	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	SHAKE	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	SHAKE	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	SHAKE	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	SHAKE	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	SHAKE	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	SHAKE	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	SHAKE	
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	SHAKE	
LCS 320-27733/2-A	Lab Control Sample	Total/NA	Solid	SHAKE	
MB 320-27733/1-A	Method Blank	Total/NA	Solid	SHAKE	

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

LCMS (Continued)

Prep Batch: 27742

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	Filtration	
320-4452-6 MS	EPAZ5S31BS3	Total/NA	Water	Filtration	
320-4452-6 MSD	EPAZ5S31BS3	Total/NA	Water	Filtration	
LCS 320-27742/2-A	Lab Control Sample	Total/NA	Water	Filtration	
MB 320-27742/1-A	Method Blank	Total/NA	Water	Filtration	

Analysis Batch: 27814

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4450-A-4-A MS	Matrix Spike	Total/NA	Water	GCMSMS_NDM A	27333
320-4450-A-4-B MSD	Matrix Spike Duplicate	Total/NA	Water	GCMSMS_NDM A	27333
320-4452-6	EPAZ5S31BS3	Total/NA	Water	GCMSMS_NDM A	27333
LCS 320-27333/2-A	Lab Control Sample	Total/NA	Water	GCMSMS_NDM A	27333
MB 320-27333/1-A	Method Blank	Total/NA	Water	GCMSMS_NDM A	27333

Analysis Batch: 27864

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-6	EPAZ5S31BS3	Total/NA	Water	6850	27626
720-52885-A-10-B MS	Matrix Spike	Total/NA	Water	6850	27626
720-52885-A-10-C MSD	Matrix Spike Duplicate	Total/NA	Water	6850	27626
LCS 320-27626/2-A	Lab Control Sample	Total/NA	Water	6850	27626
MB 320-27626/1-A	Method Blank	Total/NA	Water	6850	27626

Analysis Batch: 28015

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	6850	27733
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	6850	27733
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	6850	27733
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	6850	27733
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	6850	27733
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	6850	27733
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	6850	27733
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	6850	27733
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	6850	27733
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	6850	27733
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	6850	27733
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	6850	27733
LCS 320-27733/2-A	Lab Control Sample	Total/NA	Solid	6850	27733
MB 320-27733/1-A	Method Blank	Total/NA	Solid	6850	27733

Analysis Batch: 28138

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4421-C-3-C MSD	Matrix Spike Duplicate	Total/NA	Solid	GCMSMS_NDM A	27542
320-4421-D-3-B MS	Matrix Spike	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	GCMSMS_NDM A	27542

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

LCMS (Continued)

Analysis Batch: 28138 (Continued)

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	GCMSMS_NDM A	27542
LCS 320-27542/2-A	Lab Control Sample	Total/NA	Solid	GCMSMS_NDM A	27542
MB 320-27542/1-A	Method Blank	Total/NA	Solid	GCMSMS_NDM A	27542

Analysis Batch: 28194

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	GCMSMS_NDM A	27542
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	GCMSMS_NDM A	27542
MB 320-27642/1-A	Method Blank	Total/NA	Solid	GCMSMS_NDM A	27642

Analysis Batch: 28437

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	8321A	27706
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	8321A	27706
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	8321A	27706
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	8321A	27706
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	8321A	27706
320-4452-6	EPAZ5S31BS3	Total/NA	Water	8321A	27742
320-4452-6 MS	EPAZ5S31BS3	Total/NA	Water	8321A	27742
320-4452-6 MSD	EPAZ5S31BS3	Total/NA	Water	8321A	27742
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	8321A	27706
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	8321A	27706
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	8321A	27706
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	8321A	27706
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	8321A	27706
320-4452-11 MS	EPAZ5S31DS53	Total/NA	Solid	8321A	27706
320-4452-11 MSD	EPAZ5S31DS53	Total/NA	Solid	8321A	27706
LCS 320-27706/2-A	Lab Control Sample	Total/NA	Solid	8321A	27706
LCS 320-27742/2-A	Lab Control Sample	Total/NA	Water	8321A	27742
MB 320-27706/1-A	Method Blank	Total/NA	Solid	8321A	27706
MB 320-27742/1-A	Method Blank	Total/NA	Water	8321A	27742

TestAmerica Sacramento

QC Association Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

General Chemistry

Analysis Batch: 27131

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
320-4452-1	EPAZ5S31AS00	Total/NA	Solid	D 2216	
320-4452-1 DU	EPAZ5S31AS00	Total/NA	Solid	D 2216	
320-4452-2	EPAZ5S31AS03	Total/NA	Solid	D 2216	
320-4452-3	EPAZ5S31BS00	Total/NA	Solid	D 2216	
320-4452-4	EPAZ5S31BS03	Total/NA	Solid	D 2216	
320-4452-5	EPAZ5S31BS08	Total/NA	Solid	D 2216	
320-4452-7	EPAZ5S31CS00	Total/NA	Solid	D 2216	
320-4452-8	EPAZ5S31CS08	Total/NA	Solid	D 2216	
320-4452-9	EPAZ5S31DS00	Total/NA	Solid	D 2216	
320-4452-10	EPAZ5S31DS03	Total/NA	Solid	D 2216	
320-4452-11	EPAZ5S31DS53	Total/NA	Solid	D 2216	

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31AS00

Date Collected: 10/07/13 08:50

Date Received: 10/08/13 08:50

Lab Sample ID: 320-4452-1

Matrix: Solid

Percent Solids: 93.4

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.49 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTMS		1	4.49 g	5 mL	27878	10/18/13 15:13	WLS	TAL SAC
Total/NA	Prep	3550B			30.17 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	30.17 g	3 mL	27994	10/21/13 16:47	UFB	TAL SAC
Soluble	Leach	DI Leach			10.04 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 19:11	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 19:11	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.07 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.07 g	10.00 mL	28015	10/21/13 21:24	NSS	TAL SAC
Total/NA	Prep	3550B			9.19 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	9.19 g	0.5 mL	28138	10/23/13 06:26	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.12 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.12 g	100 mL	28437	10/26/13 00:22	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31AS03

Date Collected: 10/07/13 08:55

Date Received: 10/08/13 08:50

Lab Sample ID: 320-4452-2

Matrix: Solid

Percent Solids: 96.8

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			5.46 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTMS		1	5.46 g	5 mL	27514	10/15/13 02:03	WLS	TAL SAC
Total/NA	Prep	3550B			30.45 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	30.45 g	3 mL	27994	10/21/13 17:16	UFB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 19:22	JCB	TAL SAC
Soluble	Leach	DI Leach			10.01 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 19:22	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.06 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.06 g	10.00 mL	28015	10/21/13 21:41	NSS	TAL SAC
Total/NA	Prep	3550B			10.90 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.90 g	0.5 mL	28138	10/23/13 06:49	VMN	TAL SAC
Total/NA	Prep	SHAKE			9.99 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	9.99 g	100 mL	28437	10/26/13 00:40	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31BS00

Date Collected: 10/07/13 11:05

Date Received: 10/08/13 08:50

Lab Sample ID: 320-4452-3

Matrix: Solid

Percent Solids: 98.7

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.35 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC

TestAmerica Sacramento

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS00

Lab Sample ID: 320-4452-3

Date Collected: 10/07/13 11:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.7

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	8260B/CA_LUFTM S		1	4.35 g	5 mL	27514	10/15/13 02:27	WLS	TAL SAC
Total/NA	Prep	3550B			30.29 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	30.29 g	3 mL	27994	10/21/13 17:45	UFB	TAL SAC
Soluble	Leach	DI Leach			10.00 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 19:34	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 19:34	JCB	TAL SAC
Total/NA	Prep	SHAKE			2.99 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	2.99 g	10.00 mL	28015	10/21/13 21:58	NSS	TAL SAC
Total/NA	Prep	3550B			9.41 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	9.41 g	0.5 mL	28138	10/23/13 07:13	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.06 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.06 g	100 mL	28437	10/26/13 00:57	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31BS03

Lab Sample ID: 320-4452-4

Date Collected: 10/07/13 11:10

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.4

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.62 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	4.62 g	5 mL	27878	10/18/13 16:00	WLS	TAL SAC
Total/NA	Prep	3550B			30.52 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	30.52 g	3 mL	27994	10/21/13 18:14	UFB	TAL SAC
Soluble	Leach	DI Leach			10.04 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 19:45	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 19:45	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.12 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.12 g	10.00 mL	28015	10/21/13 22:14	NSS	TAL SAC
Total/NA	Prep	3550B			10.42 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.42 g	0.5 mL	28138	10/23/13 07:37	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.08 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.08 g	100 mL	28437	10/26/13 01:15	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31BS08

Lab Sample ID: 320-4452-5

Date Collected: 10/07/13 11:35

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 97.7

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			5.47 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	5.47 g	5 mL	27878	10/18/13 16:23	WLS	TAL SAC

TestAmerica Sacramento

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31BS08

Lab Sample ID: 320-4452-5

Date Collected: 10/07/13 11:35

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 97.7

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3550B			29.28 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.28 g	3 mL	27994	10/21/13 18:43	UFB	TAL SAC
Soluble	Leach	DI Leach			10.07 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 19:57	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 19:57	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.02 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.02 g	10.00 mL	28015	10/21/13 22:31	NSS	TAL SAC
Total/NA	Prep	3550B			10.26 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.26 g	0.5 mL	28138	10/23/13 08:00	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.09 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.09 g	100 mL	28437	10/26/13 01:32	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31BS3

Lab Sample ID: 320-4452-6

Date Collected: 10/07/13 13:30

Matrix: Water

Date Received: 10/08/13 08:50

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Analysis	8260B/CA_LUFTMS		1	50 mL	50 mL	27338	10/11/13 20:55	WLS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTMS	RA	1	50 mL	50 mL	28211	10/22/13 18:05	KSS	TAL SAC
Total/NA	Prep	3510C			1044.7 mL	3 mL	27435	10/14/13 10:08	NGK	TAL SAC
Total/NA	Analysis	8015B		1	1044.7 mL	3 mL	27994	10/22/13 06:20	UFB	TAL SAC
Total/NA	Analysis	9056		1	5 mL		28225	10/08/13 14:08	LW1	TAL SAC
Total/NA	Analysis	9056		1	5 mL		28226	10/08/13 14:08	LW1	TAL SAC
Total/NA	Prep	3510C			1055.2 mL	0.5 mL	27333	10/11/13 12:19	AVM	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	1055.2 mL	0.5 mL	27814	10/18/13 06:25	VMN	TAL SAC
Total/NA	Prep	Filtration			4.00 mL	4.00 mL	27626	10/16/13 07:14	HJA	TAL SAC
Total/NA	Analysis	6850		1	4.00 mL	4.00 mL	27864	10/19/13 09:01	NSS	TAL SAC
Total/NA	Prep	Filtration			10.00 mL	10.00 mL	27742	10/17/13 08:50	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.00 mL	10.00 mL	28437	10/25/13 21:44	DEG	TAL SAC

Client Sample ID: EPAZ5S31CS00

Lab Sample ID: 320-4452-7

Date Collected: 10/07/13 10:15

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.51 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTMS		1	4.51 g	5 mL	27878	10/18/13 16:47	WLS	TAL SAC
Total/NA	Prep	3550B			29.93 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.93 g	3 mL	27994	10/21/13 19:12	UFB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 20:08	JCB	TAL SAC

TestAmerica Sacramento

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31CS00

Lab Sample ID: 320-4452-7

Date Collected: 10/07/13 10:15

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Soluble	Leach	DI Leach			10.01 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 20:08	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.04 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.04 g	10.00 mL	28015	10/21/13 22:48	NSS	TAL SAC
Total/NA	Prep	3550B			9.60 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	9.60 g	0.5 mL	28138	10/23/13 08:24	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.07 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.07 g	100 mL	28437	10/26/13 01:50	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31CS08

Lab Sample ID: 320-4452-8

Date Collected: 10/07/13 10:55

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 98.8

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			5.12 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	5.12 g	5 mL	27878	10/18/13 17:10	WLS	TAL SAC
Total/NA	Prep	3550B			29.20 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.20 g	3 mL	27994	10/21/13 19:41	UFB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 20:20	JCB	TAL SAC
Soluble	Leach	DI Leach			10.13 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 20:20	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.00 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.00 g	10.00 mL	28015	10/21/13 23:05	NSS	TAL SAC
Total/NA	Prep	3550B			10.74 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.74 g	0.5 mL	28138	10/23/13 08:51	VMN	TAL SAC
Total/NA	Prep	SHAKE			9.99 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	9.99 g	100 mL	28437	10/26/13 02:08	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31DS00

Lab Sample ID: 320-4452-9

Date Collected: 10/07/13 07:20

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.95 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	4.95 g	5 mL	27878	10/18/13 17:33	WLS	TAL SAC
Total/NA	Prep	3550B			29.54 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.54 g	3 mL	27994	10/21/13 20:10	UFB	TAL SAC
Soluble	Leach	DI Leach			10.02 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 20:31	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 20:31	JCB	TAL SAC

TestAmerica Sacramento

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31DS00

Lab Sample ID: 320-4452-9

Date Collected: 10/07/13 07:20

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.6

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	SHAKE			3.01 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.01 g	10.00 mL	28015	10/21/13 23:21	NSS	TAL SAC
Total/NA	Prep	3550B			9.62 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	9.62 g	0.5 mL	28194	10/23/13 13:33	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.05 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.05 g	100 mL	28437	10/26/13 02:25	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31DS03

Lab Sample ID: 320-4452-10

Date Collected: 10/07/13 08:05

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 99.3

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			4.87 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	4.87 g	5 mL	27878	10/18/13 17:57	WLS	TAL SAC
Total/NA	Prep	3550B			29.29 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.29 g	3 mL	27994	10/21/13 22:07	UFB	TAL SAC
Soluble	Leach	DI Leach			10.10 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 21:06	JCB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 21:06	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.02 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.02 g	10.00 mL	28015	10/21/13 23:38	NSS	TAL SAC
Total/NA	Prep	3550B			10.82 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.82 g	0.5 mL	28194	10/23/13 13:56	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.06 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.06 g	100 mL	28437	10/26/13 03:00	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Client Sample ID: EPAZ5S31DS53

Lab Sample ID: 320-4452-11

Date Collected: 10/07/13 08:30

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	5035			5.5 g	5 mL	27074	10/08/13 17:53	KSS	TAL SAC
Total/NA	Analysis	8260B/CA_LUFTM S		1	5.5 g	5 mL	27514	10/15/13 01:17	WLS	TAL SAC
Total/NA	Prep	3550B			29.74 g	3 mL	27563	10/15/13 11:23	ALH	TAL SAC
Total/NA	Analysis	8015B		1	29.74 g	3 mL	27994	10/21/13 22:36	UFB	TAL SAC
Soluble	Analysis	9056		1	5 mL		28765	10/24/13 21:17	JCB	TAL SAC
Soluble	Leach	DI Leach			10.11 g	50 mL	28360	10/24/13 10:30	LW1	TAL SAC
Soluble	Analysis	9056		1	5 mL		28766	10/24/13 21:17	JCB	TAL SAC
Total/NA	Prep	SHAKE			3.09 g	10.00 mL	27733	10/17/13 06:52	HJA	TAL SAC
Total/NA	Analysis	6850		1	3.09 g	10.00 mL	28015	10/21/13 23:55	NSS	TAL SAC

TestAmerica Sacramento

Lab Chronicle

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Client Sample ID: EPAZ5S31DS53

Lab Sample ID: 320-4452-11

Date Collected: 10/07/13 08:30

Matrix: Solid

Date Received: 10/08/13 08:50

Percent Solids: 96.8

Prep Type	Batch Type	Batch Method	Run	Dil Factor	Initial Amount	Final Amount	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3550B			10.51 g	0.5 mL	27542	10/15/13 10:04	ALH	TAL SAC
Total/NA	Analysis	GCMSMS_NDMA		1	10.51 g	0.5 mL	28194	10/23/13 14:20	VMN	TAL SAC
Total/NA	Prep	SHAKE			10.00 g	100 mL	27706	10/16/13 13:43	HJA	TAL SAC
Total/NA	Analysis	8321A		1	10.00 g	100 mL	28437	10/26/13 03:18	DEG	TAL SAC
Total/NA	Analysis	D 2216		1			27131	10/09/13 13:23	LW1	TAL SAC

Laboratory References:

TAL SAC = TestAmerica Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Certification Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Laboratory: TestAmerica Sacramento

All certifications held by this laboratory are listed. Not all certifications are applicable to this report.

Authority	Program	EPA Region	Certification ID	Expiration Date
A2LA	A2LA		NE-OS-22-13	01-31-14
A2LA	DoD ELAP		2928-01	01-31-14
Alaska (UST)	State Program	10	UST-055	12-18-13
Arizona	State Program	9	AZ0708	08-11-14
Arkansas DEQ	State Program	6	88-0691	06-17-14
California	NELAP	9	1119CA	01-31-14
Connecticut	State Program	1	PH-0691	06-30-15
Florida	NELAP	4	E87570	06-30-14
Guam	State Program	9	N/A	08-31-14
Hawaii	State Program	9	N/A	01-31-14
Illinois	NELAP	5	200060	03-17-14
Kansas	NELAP	7	E-10375	10-31-14
Louisiana	NELAP	6	30612	06-30-14
Michigan	State Program	5	9947	01-31-14
Nebraska	State Program	7	NE-OS-22-13	01-31-14
Nevada	State Program	9	CA44	07-31-14
New Jersey	NELAP	2	CA005	06-30-14
New York	NELAP	2	11666	04-01-14
Northern Mariana Islands	State Program	9	MP0007	02-01-14
Oregon	NELAP	10	CA200005	03-28-14
Pennsylvania	NELAP	3	68-01272	03-31-14
South Carolina	State Program	4	87014	06-30-14
Texas	NELAP	6	T104704399-08-TX	05-31-14
US Fish & Wildlife	Federal		LE148388-0	12-31-13
USDA	Federal		P330-11-00436	12-30-14
USEPA UCMR	Federal	1	CA00044	11-06-14
Utah	NELAP	8	QUAN1	01-31-14
Washington	State Program	10	C581	05-05-14
West Virginia	State Program	3	9930C	12-31-13
Wyoming	State Program	8	8TMS-Q	01-31-14

TestAmerica Sacramento

Method Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Method	Method Description	Protocol	Laboratory
8260B/CA_LUFTM S	Volatile Organic Compounds by GC/MS	SW846	TAL SAC
8015B	Diesel Range Organics (DRO) (GC)	SW846	TAL SAC
9056	Anions, Ion Chromatography	SW846	TAL SAC
6850	Perchlorate by LC/MS or LC/MS/MS	EPA	TAL SAC
8321A	Benzene Sulfonic Acids (LCMS)	SW846	TAL SAC
GCMSMS_NDMA	Nitrosamines by Isotope Dilution and GC/CI/MS/MS	TestAmerica SOP	TAL SAC
D 2216	Percent Moisture	ASTM	TAL SAC

Protocol References:

ASTM = ASTM International

EPA = US Environmental Protection Agency

SW846 = "Test Methods For Evaluating Solid Waste, Physical/Chemical Methods", Third Edition, November 1986 And Its Updates.

TestAmerica SOP = TestAmerica, Inc., Standard Operating Procedure

Laboratory References:

TAL SAC = TestAmerica Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Sample Summary

Client: CH2M Hill Constructors, Inc.
Project/Site: Stringfellow Acid Project

TestAmerica Job ID: 320-4452-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
320-4452-1	EPAZ5S31AS00	Solid	10/07/13 08:50	10/08/13 08:50
320-4452-2	EPAZ5S31AS03	Solid	10/07/13 08:55	10/08/13 08:50
320-4452-3	EPAZ5S31BS00	Solid	10/07/13 11:05	10/08/13 08:50
320-4452-4	EPAZ5S31BS03	Solid	10/07/13 11:10	10/08/13 08:50
320-4452-5	EPAZ5S31BS08	Solid	10/07/13 11:35	10/08/13 08:50
320-4452-6	EPAZ5S31BS3	Water	10/07/13 13:30	10/08/13 08:50
320-4452-7	EPAZ5S31CS00	Solid	10/07/13 10:15	10/08/13 08:50
320-4452-8	EPAZ5S31CS08	Solid	10/07/13 10:55	10/08/13 08:50
320-4452-9	EPAZ5S31DS00	Solid	10/07/13 07:20	10/08/13 08:50
320-4452-10	EPAZ5S31DS03	Solid	10/07/13 08:05	10/08/13 08:50
320-4452-11	EPAZ5S31DS53	Solid	10/07/13 08:30	10/08/13 08:50

**STRINGFELLOW ZONE #5
CH2MHILL**

**6 HUTTON CENTRE DRIVE, SUITE 700
SANTA ANA, CA 92707
CONTACT: EMILY KEENE
CONTACT PHONE: 714-435-6097**

00713-140020-0095
erica Laboratories Inc.
Lab Contact: Jill Kellman
Lab Phone: 916-374-4402



320-4452 Chain of Custody

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USEPA CLP Generic COC (LAB COPY)

Date Shipped: 10/7/2013

Carrier Name: FedEx

Airbill No: 796741820318/796763189279

Sample Identifier	CLP Sample No.	Matrix/Sampler	Coll. Method	Analysis/Turnaround (Days)	Tag/Preservation	Location	Collection Date/Time	For Lab Use Only
EPAZ5S31AS00	EPAZ5S31A S00	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31A	10/07/2013 08:50	
EPAZ5S31AS03	EPAZ5S31A S03	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31A	10/07/2013 08:55	
EPAZ5S31BS00	EPAZ5S31B S00	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31B	10/07/2013 11:05	

Special Instructions:	Shipment for Case Complete? N
Analysis Key	Samples Transferred From Chain of Custody #

Items/Reason	Relinquished by (Signature and Organization)	Date/Time	Received by (Signature and Organization)	Date/Time	Sample Condition Upon Receipt
		10/7/13 1500		10/9/13 850	3.2°C, 1.7°C

STRINGFELLOW ZONE #5

CH2MHILL

6 HUTTON CENTRE DRIVE, SUITE 700

SANTA ANA, CA 92707

CONTACT: EMILY KEENE

CONTACT PHONE: 714-435-6097

No: 9-100713-140020-0095

Lab: TestAmerica Laboratories Inc.

Lab Contact: Jill Kellman

Lab Phone: 916-374-4402

USEPA CLP Generic COC (LAB COPY)

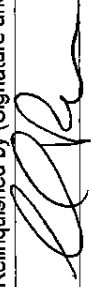
Date Shipped: 10/7/2013

Carrier Name: FedEx

Airbill No: 796741820318/796763189279

Sample Identifier	CLP Sample No.	Matrix/Sampler	Coll. Method	Analysis/Turnaround (Days)	Tag/Preservative/Bottles	Location	Collection Date/Time	For Lab Use Only
EPAZ5S31BS03	EPAZ5S31B S03	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31B	10/07/2013 11:10	
EPAZ5S31BS08	EPAZ5S31B S08	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31B	10/07/2013 11:35	
EPAZ5S31BS3	EPAZ5S31B S3	WATER/ Mike Ladeau	Grab	NDMA LL (WS-MS-0012r1.5)(15), Perchlorate (EPA 6850)(15), GRO (CA LUFT)(15), DRO (CA LUFT)(15), pCBA (EPA 8321) + Anions (EPA 300.0)(15)	A (4 C), B (4 C), E (4 C / HCL), G (4 C), T (4 C) (8)	EPAZ5S31B	10/07/2013 13:30	

Special Instructions:	Shipment for Case Complete? N
Analysis Key	Samples Transferred From Chain of Custody #

Items/Reason	Relinquished by (Signature and Organization)	Date/Time	Received by (Signature and Organization)	Date/Time	Sample Condition Upon Receipt
		10/7/13 1:50		10/13/13 8:50	

**STRINGFELLOW ZONE #5
CH2MHILL**

**6 HUTTON CENTRE DRIVE, SUITE 700
SANTA ANA, CA 92707
CONTACT: EMILY KEENE
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No: 9-100713-140020-0095
Lab: TestAmerica Laboratories Inc.
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USEPA CLP Generic COC (LAB COPY)

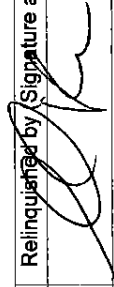
Date Shipped: 10/7/2013

Carrier Name: FedEx

Airbill No: 796741820318/796763189279

Sample Identifier	CLP Sample No.	Matrix/Sampler	Coll. Method	Analysis/Turnaround (Days)	Tag/Preservative/Bottles	Location	Collection Date/Time	For Lab Use Only
EPAZ5S31CS00	EPAZ5S31C S00	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), PCB SA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31C	10/07/2013 10:15	
EPAZ5S31CS03	EPAZ5S31C S03	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), PCB SA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31C	10/07/2013 10:20	
EPAZ5S31CS08	EPAZ5S31C S08	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), PCB SA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31C	10/07/2013 10:55	

Special Instructions:	Shipment for Case Complete? N
Analysis Key	Samples Transferred From Chain of Custody #

Items/Reason	Relinquished by Signature and Organization	Date/Time	Received by (Signature and Organization)	Date/Time	Sample Condition Upon Receipt
		10/7/13 1500		10/8/13 8:50	

did not receive or 10-8-13

USEPA CLP Generic COC (LAB COPY)

Date Shipped: 10/7/2013

Carrier Name: FedEx

Airbill No: 796741820318/796763189279

CHAIN OF CUSTODY RECORD

Project Code:

Cooler #: 2

No: 9-100713-140020-0095

Lab: TestAmerica Laboratories Inc.

Lab Contact: Jill Kellman

Lab Phone: 916-374-4402

Sample Identifier	CLP Sample No.	Matrix/Sampler	Coll. Method	Analysis/Turnaround (Days)	Tag/Preservative/Bottles	Location	Collection Date/Time	For Lab Use Only
EPAZ5S31DS00	EPAZ5S31D S00	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31D	10/07/2013 07:20	
EPAZ5S31DS03	EPAZ5S31D S03	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (5)	EPAZ5S31D	10/07/2013 08:05	
EPAZ5S31DS53	EPAZ5S31D S58	SOIL/ Mike Ladeau	Hand Auger	NDMA LL (WS-MS-0012r1.5) + Perch (EPA 6850)(15), GRO (CA LUFT)(15), pCBSA (EPA8321) + Anions (EPA9056) + DRO (CALUFT)(15)	Q (4 C), R (4 C), T (4 C) (12)	EPAZ5S31D	10/07/2013 08:30	

Shipment for Case Complete? N

Samples Transferred From Chain of Custody #

Sample(s) to be used for Lab QC: EPAZ5S31DS53 Tag Q, EPAZ5S31DS53 Tag R, EPAZ5S31DS53 Tag T

Analysis Key

Items/Reason	Relinquished by (Signature and Organization)	Date/Time	Received by (Signature and Organization)	Date/Time	Sample Condition Upon Receipt
		10/7/13 1500		10/8/13 8:50	

Login Sample Receipt Checklist

Client: CH2M Hill Constructors, Inc.

Job Number: 320-4452-1

Login Number: 4452

List Source: TestAmerica Sacramento

List Number: 1

Creator: Hytrek, Cheryl

Question	Answer	Comment
Radioactivity wasn't checked or is $\leq$ background as measured by a survey meter.	True	
The cooler's custody seal, if present, is intact.	True	
Sample custody seals, if present, are intact.	N/A	
The cooler or samples do not appear to have been compromised or tampered with.	True	
Samples were received on ice.	True	
Cooler Temperature is acceptable.	True	
Cooler Temperature is recorded.	True	
COC is present.	True	
COC is filled out in ink and legible.	True	
COC is filled out with all pertinent information.	True	
Is the Field Sampler's name present on COC?	N/A	
There are no discrepancies between the containers received and the COC.	False	IDs on containers do not match the COC. Logged in per COC.
Samples are received within Holding Time.	True	
Sample containers have legible labels.	True	
Containers are not broken or leaking.	True	
Sample collection date/times are provided.	True	
Appropriate sample containers are used.	True	
Sample bottles are completely filled.	True	
Sample Preservation Verified.	N/A	
There is sufficient vol. for all requested analyses, incl. any requested MS/MSDs	True	
Containers requiring zero headspace have no headspace or bubble is $<6\text{mm}$ (1/4").	True	
Multiphasic samples are not present.	True	
Samples do not require splitting or compositing.	True	
Residual Chlorine Checked.	N/A	